Sec: SR.IIT_*CO-SC(MODEL-A& APEX) Date:02-01-2025

Time: 3HRS GTM-9 Max. Marks: 300

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
B	B	A	A	B	B	C	A	D	A
11	12	13	14	15	16	17	18	19	20
C	A	D	C	A	B	A	C	B	C
21	22	23	24	25					
45	20	3	2	6					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
B	A	C	D	B	D	C	C	B	B
36	37	38	39	40	41	42	43	44	45
A	C	B	A	B	A	A	B	C	B
46	47	48	49	50					
8	14	4	8	7					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
C	A	A	C	B	B	B	A	B	A
61	62	63	64	65	66	67	68	69	70
C	A	A	A	A	A	D	C	C	D
71	72	73	74	75					
0	2	2	114	14					

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PHYSICS

1. Here $y=\overline{(A.B).\overline{C}}=\overline{AB}+\overline{\overline{C}}=\overline{A}+\overline{B}+C$

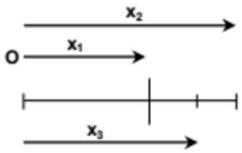
$y=0$ if $A=1, B=1$ and $C=0$

2.

$x=t^3-4t^2+5t+9$

$V=\frac{dx}{dt}=\left(3t^2-8t+5\right)$

V = 0 at t = 1, $\frac{5}{3}$ sec.



x_1 (t = 0) = 9m; x_2 (t = 1) = 11 m, x_3 (t = 5/3) ≈ 10.85

d = $(x_2 - x_2)$ + $(x_2 - x_3)$ = 2.15 m

3. energy of photon $E=\frac{1242}{\lambda(in\ nm)}eV=\frac{1242}{200}=6.2eV$

$E=\phi+K_{max}$

$6.2=4.7+K_{max}$

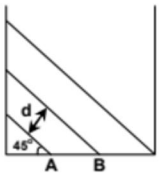
$K_{max}=1.5eV=eVs$

Stopping potential $Vs=1.5V$

The electron will be ejected until potential of sphere equals to stopping potential

$V=Vs=1.5V$

4.



$|P_A-P_B|=d\times g_{effective}\times\rho=10^4\ N/m^2$

5. $\frac{1}{\lambda_0}\propto\left[\frac{1}{2^2}-\frac{1}{3^2}\right]\Rightarrow\frac{1}{\lambda_0}\propto\frac{5}{36}\rightarrow(1)$

$\frac{1}{\lambda}\propto\left[\frac{1}{2^2}-\frac{1}{4^2}\right]\Rightarrow\frac{1}{\lambda}\propto\frac{3}{16}\rightarrow(2)$

$\frac{eq.1}{eq.2}\Rightarrow\frac{\lambda}{\lambda_0}=\frac{20}{27}\Rightarrow\lambda=\frac{20}{27}\lambda_0$

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6. Conceptual

7. $TE_{surface} = TE_{\max height}$

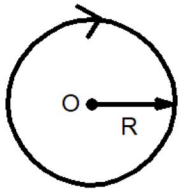
$$\frac{-GMm}{R} + \frac{1}{2}m\left(\frac{V_e}{2}\right)^2 = \frac{-GMm}{R+h}$$

$$\frac{-GMm}{R} + \frac{1}{2}m\frac{2GM}{4R} = \frac{-GMm}{R+h}$$

$$V_e^2 = \frac{2GM}{R}$$

On simplifying we get $h = \frac{R}{3}$

8.



Magnetic field at centre $B = \frac{\mu_0 i}{2R} \Rightarrow i = \frac{2RB}{\mu_0}$

Magnetic moment of loop

$$M = iA = i\pi R^2 = \frac{2RB}{\mu_0}\pi R^2 = \frac{2BR^3}{\frac{\mu_0}{\pi}} = 3\times10^{-2}\ Am^2$$

9. $E = \frac{1}{2}mA^2\omega^2$ i.e. $E \propto (A\omega)^2$

or $(A_1\omega_1)^2 = (A_2\omega_2)^2$

10. $PV^{\frac{2}{3}} = \text{constant}, \quad PV = nRT$

$$P = \frac{nRT}{V}$$

$$\frac{nRT}{V}.V^{\frac{2}{3}} = \text{constant}$$

$$TV^{-\frac{1}{3}} = \text{constant}$$

As volume ↑, temperature ↑

11. $E = \overline{V} \times \overline{B} \Rightarrow$

12. $\frac{EB}{\mu} = I\,(Intensity\ of\ EMW)$

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13. By KVL & KCL

14. $V_{rms} = \left(1+\frac{1}{x}\right)^{\frac{1}{2}} Vare$

$$\sqrt{\frac{3RT}{M}} = \left(1+\frac{1}{x}\right)^{\frac{1}{2}} \sqrt{\frac{8RT}{\pi M}}$$

$$\frac{22\times 3}{8\times 7} = 1+\frac{1}{x}$$

15. $A = \delta_m = 60°$

At minimum deviation $i = \left(\frac{A+\delta_m}{2}\right) = 60°$

16. 10 VSD=8 MSD

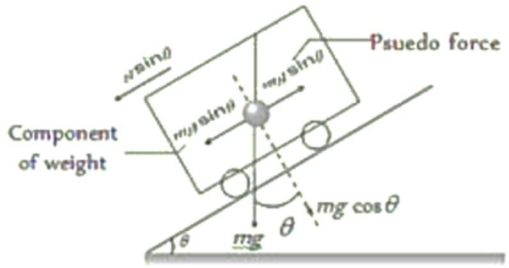
$\therefore$ 1VSD = 1 MSD-1VSD

Now LC=1 MSD – 1 VSD

= MSD – 0.8 MSD

$$= 0.2\,MSD = 0.2\times\frac{1}{10}cm = 0.02\,cm$$

17.



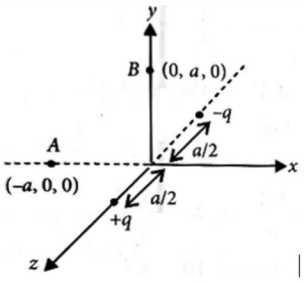
F_{net} = mg cos θ or net acceleration of the bob is g_{eff} = g cos θ

$\therefore$ Time period T = $2\pi\sqrt{\frac{1}{g_{eff}}} = 2\pi\sqrt{\frac{1}{g\cos\theta}}$

18. It can be seen that potential both at A and B are zero. When the charge is moved from A to B, workdone by the electric field on the charge will be zero

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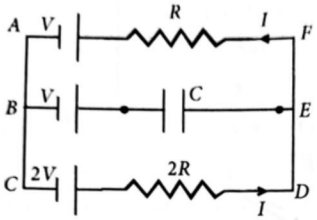
19. When a disc rolls, it has two types of motion. It possesses translational and rotational motion. For each type of motion, there is an angular momentum. Let L_T and L_R represent them.

$\therefore L = L_T + L_R \; or \; L = (Mv \times R) + (I_C \omega)$

Where C denotes centre of mass

Or $L = M(R\omega)R + \dfrac{MR^2}{2}\omega \; or \; L = \dfrac{3}{2}MR^2\omega$

20. In the steady state condition, no current flows through the capacitor C along the branch BE.



Apply Kirchhoff's law to the closed mesh ACDFA,

$2V - I(2R) - I(R) - V = 0 \; or \; V = 3IR$

Or $I = \dfrac{V}{3R}$

Apply Kirchhoff's law to the mesh ABEFA, $V + V_C - IR - V = 0 \; or \; V_C = IR = \left(\dfrac{V}{3R}\right)R = \dfrac{V}{3}$

$\therefore$ Potential difference across capacitor $V_C = \dfrac{V}{3}$

21. $\delta = (i_1 + i_2) - A$

$\therefore \quad A = (i_1 + i_2) - \delta$

$= (15° + 60°) - 30° = 45°$

22. Condition of balanced bridge.

23. Here $m = 1\,kg, \vec{F} = \left(\alpha t\,\hat{i} + \beta\,\hat{j}\right) = \left(t\,\hat{i} + \hat{j}\right)N$

As, $\vec{F} = m\dfrac{d\vec{v}}{dt} = \left(t\,\hat{i} + \hat{j}\right) \; or \; \vec{v} = \int_0^t \left(t\,\hat{i} + \hat{j}\right)dt$

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$$\vec{v}=\left[\frac{t^2}{2}\hat{i}+t\,\hat{j}\right]$$

Also, $\vec{r}=\int_0^t \vec{v}\,dt=\int_0^t\left(\frac{t^2}{2}\hat{i}+t\,\hat{j}\right)dt$; $\vec{r}\left(\frac{t^3}{6}\hat{i}+\frac{t^2}{2}\hat{j}\right)$

Velocity of the body at $t=1\,s$,

$$\vec{v}=\left(\frac{1}{2}\hat{i}+\hat{j}\right)ms^{-1}=\frac{1}{2}\left(\hat{i}+2\,\hat{j}\right)ms^{-1}$$

Displacement of the body at $t=1\,s$,

$$\vec{s}=\vec{r}(t=1s)-\vec{r}(t=0\,s)$$

$$=\left(\frac{1}{6}\hat{i}+\frac{1}{2}\hat{j}\right)-0=\frac{1}{6}\left(\hat{i}+3\,\hat{j}\right)m$$

$$\left|\vec{s}\right|=\frac{1}{6}\sqrt{1^2+3^2}=\frac{\sqrt{10}}{6}$$

Torque on the body at $t=1\,s$,

$$\vec{\tau}=\vec{r}\times\vec{F}=\left(\frac{1}{6}\hat{i}+\frac{1}{2}\hat{j}\right)\times\left(\hat{i}+\hat{j}\right)$$

$$=\frac{1}{6}\left(\hat{i}\times\hat{j}\right)+\frac{1}{2}\left(\hat{j}\times\hat{i}\right)=\frac{1}{6}\hat{k}-\frac{1}{2}\hat{k}=-\frac{1}{3}\hat{k}$$

$$\left|\vec{\tau}\right|=\frac{1}{3}N\,m$$

24. For circle $2\pi R=L$ *or* $R=\frac{L}{2\pi}$

$\therefore$ Magnetic moment $=iA=i\times\pi R^2$ $\therefore M=\frac{L^2 i}{4\pi}$

25. The excess of pressure above atmospheric pressure, due to surface tension in a bubble $=\frac{4T}{r}$

The surrounding pressure $=P_0+\frac{4T}{r_A}=8+\frac{4\times0.04}{0.02}$

$P_A=16N/m^2$

$$P_B=P_0+\frac{4T}{r_B}=8+\frac{4\times0.04}{0.04}=12\ N/m^2$$

$PV=nRT$

$$(16)\frac{4}{3}\pi(0.02)^3=n_A RT$$

$$(12)\left(\frac{4}{3}\pi(0.04)^3\right)=n_B RT$$

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$$\frac{n_B}{n_A} = 6$$

CHEMISTRY

26. $P^0 \propto 2.05$

$P_s \propto 2.0$

$$\frac{P^0 - P_s}{P_s} = \frac{n_B}{n_A} = \frac{5/m_B}{90/18} \Rightarrow \frac{0.05}{2} = \frac{1}{m_B} \Rightarrow m_B = 40$$

27.

$$2H_2(g) \longrightarrow 4H^+ + 4e^- \quad E° = 0.0V$$

$$O_2(g) + 4H^+(aq) + 4e^- \longrightarrow 2H_2O(l) \quad E° = 1.23V$$

$$2H_2(g) + O_2(g) \longrightarrow 2H_2O(l) \quad E°_{cell} = 1.23V$$

$$\Delta G°_{298} = -nFE° = -4 \times 96500 \times 1.23 = -474.78kJ$$

$$\Delta G°_{298} = \Delta H°_{298} - T.\Delta S°_{298}$$

$$\Delta S°_{298} = \frac{2 \times (-285.5) - (-475)}{298} = -0.322\,kJ/K$$

28. As per common ion effect

29.

	$N_2O_5(g) \rightleftharpoons 2NO_2(g) + \frac{1}{2}O_2(g)$		
At $t=0$	a	0	0
At $t=t_{50}$	$a-x$	$2x$	$\frac{1}{2}x$
At $t=t_{100}$	$a-y$	$2y$	$\frac{1}{2}y$

$a = 50mmHg$

$$a + \frac{3}{2}x = 87.5$$

$x = 25$

$y = 37.5$

For first order reaction, $kt = 2.303\log\left(\frac{a}{a-x}\right)$

At $50\,\text{min}$, $kt = 2.303\log\left(\frac{50}{50-25}\right)$

$kt = 2.303\log 2$

$$\Rightarrow k = \frac{2.303 \times 0.3010}{50}$$

$kt = 2.303\log 2$

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$$\frac{50}{a-y}=4$$

$$a-y=\frac{50}{4}=12.5$$

$50-y=12.5 \Rightarrow y=37.5$

Therefore, total pressure at 100 min can be Calculated as

Total pressure $=a-y+2y+\frac{1}{2}y$

30.

$A_2B_3(aq) \longrightarrow 2A^{3+}(aq)+3B^{2-}(aq)$

$n=5$

$so, i=1+(5-1)\times\alpha$

$\Delta T_b = iK_b m = (1+4\alpha)K_b m$

$=(1+4\times0.7)\times0.5\times1=1.9; \quad T_b=373+1.9$

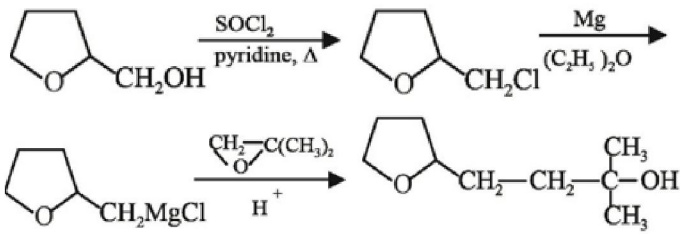
31.

$4HCl(g) \longrightarrow 2H_2(g)+2Cl_2(g) \quad \Delta_r H^\circ = 2\times184.5$

$2H_2(g)+O_2(g) \longrightarrow 2H_2O(g) \quad \Delta_r H^\circ = -483$

$4HCl(g)+O_2(g) \longrightarrow 2Cl_2(g)+2H_2O(g) \quad \Delta_r H^\circ = -114$

32.



33. Let volume of butane = x ml

Volume of propane = y ml

Volume of propyne = zml

x+y+z=150

Given x=60

$\therefore$ y+z=90

x ml of C_4H_{10} produce 4x ml CO_2

y ml of C_3H_8 produce 3y ml CO_2

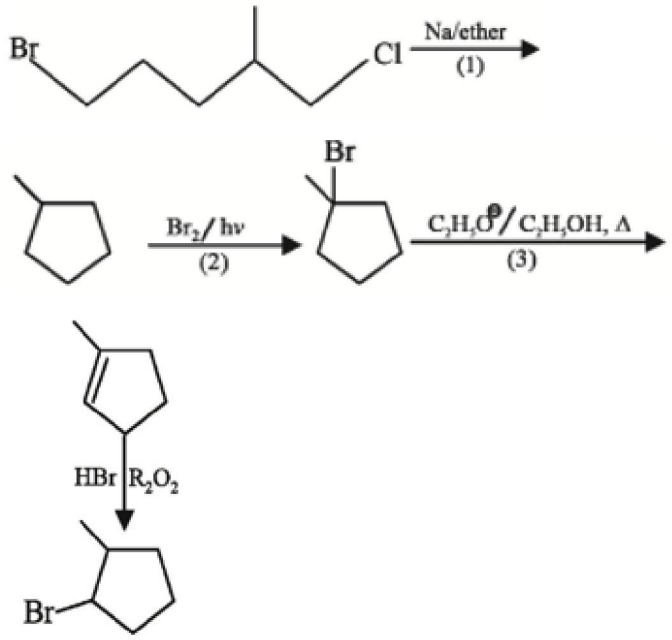
z ml of C_3H_4 produce 3z ml CO_2

$\therefore$ 150 ml mixture will produce $=4x+3(y+z)=4\times60+3\times90=510ml$

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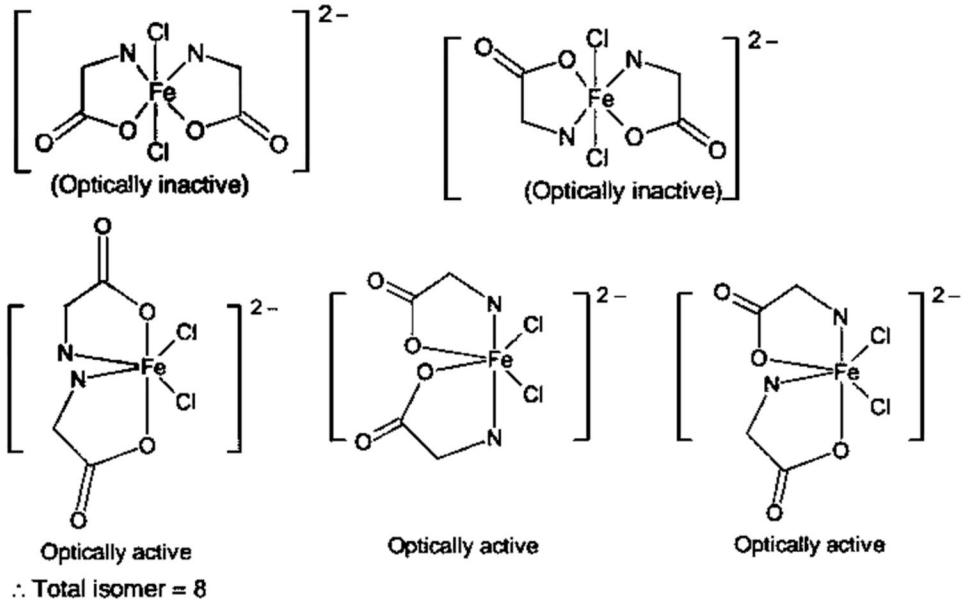
34.



35. –M group destabilizes the carbocation therefore slowest rate of electrophilic addition.

36. $[MnO_4]^{2-}$ Mn is in +6 oxidation state. Electronic configuration is $[Ar]^{18}\,3d^1 4s^0$ As it contains one unpaired electron it is paramagnetic and green coloured because of d-d transition in visible region. $[TiCl_6]^{2-}$, $[VO_4]^{3-}$ and CrO_2Cl_2 are diamagnetic because all electrons are paired.

37.

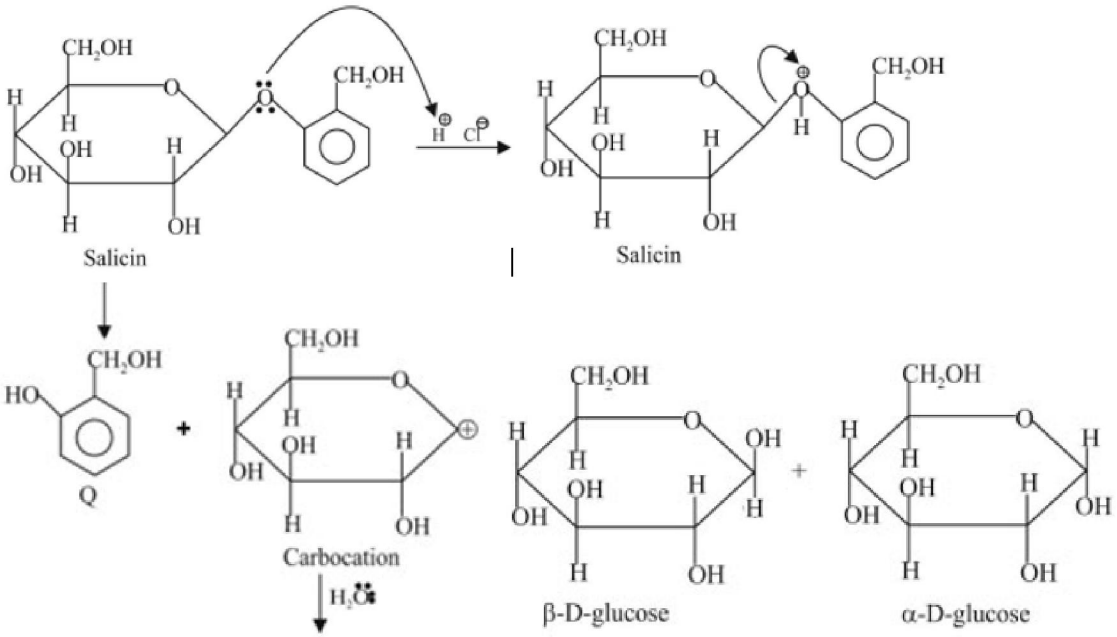


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38. Reduction with $SnCl_2$+HCl is Stephen's reduction.

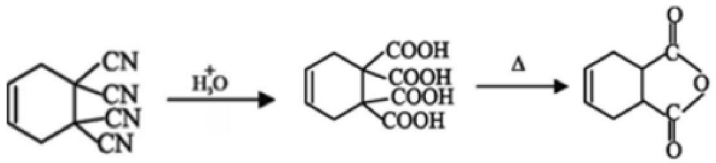
39. Conceptual

40.



41. For isoelectronic species ionic radii increase as the nuclear charge decrease. So correct order is Mg^{2+}, Na^{+}, F^{-}, O^{2-}, N^{3-}

42.



43. EN values

C	Si	Ge	Sn	Pb
2.5	1.8	1.8	1.8	1.9

44.

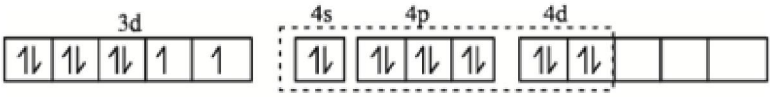
Ion	Electronic configuration
Eu^{2+}	$4f^7$
Tm^{2+}	$4f^{13}$
Sm^{2+}	$4f^6$
Tm^{3+}	$4f^{12}$
Tb^{4+}	$4f^7$
Yb^{2+}	$4f^{14}$
Dy^{3+}	$4f^9$
Yb^{3+}	$4f^{13}$

45. NH_3 is strong field ligand and may pair up e^{-} of metal if there is any benefit and hybridization becomes d^2sp^3 otherwise no pairing is done

$_{28}Ni$ is in +2 oxidation state with electron configuration $[Ar]^{18}\,3d^8\,4s^0$, so $\left[Ni(NH_3)_6\right]^{2+}$

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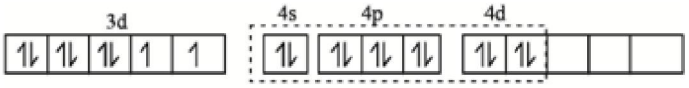
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sp^3d^2 hybridisation

a)

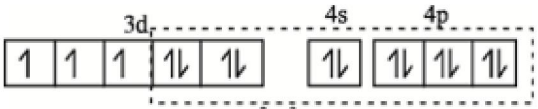
Number of unpaired electrons = 2



sp^3d^2 hybridisation

b)

$_{24}Cr$ is in +3 oxidation state with electron configuration $[Ar]^{18}\,3d^3\,4s^0$; so $\left[Cr(NH_3)_6\right]^{3+}$

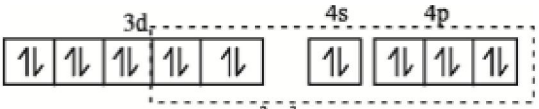


d^2sp^3 hybridisation

Number of unpaired electrons remains three, whether ligand is strong field or weak ligand when electron configuration is $3d^3$.

c) $_{27}Co$ is in +3 oxidation state with electron configuration

$[Ar]^{18}\,3d^6\,4s^0$, so $\left[Co(NH_3)_6\right]^{3+}$

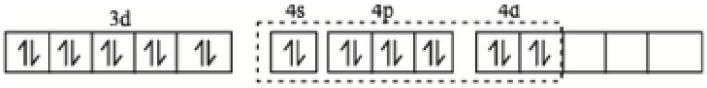


d^2sp^3 hybridisation

All electrons are paired, so complex ion is diamagnetic.

d) $_{30}Zn$ is in +2 oxidation state with electron configuration

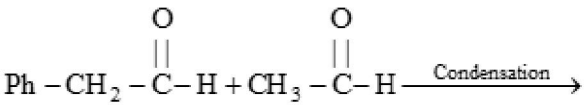
$[Ar]^{18}\,3d^{10}\,4s^0$, so $[Zn(NH_3)_6]^{2+}$

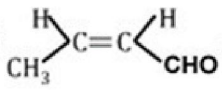
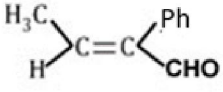
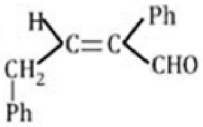
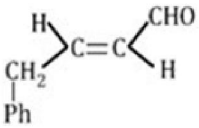


sp^3d^2 hybridisation

All electrons are paired, so complex ion is diamagnetic.

46.



(1)  (2)  (3)  (4) 

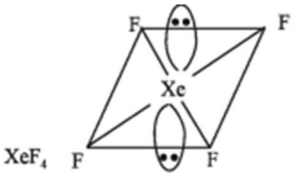
All four having geometrical isomers each.

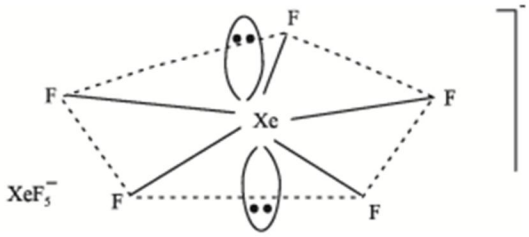
47.

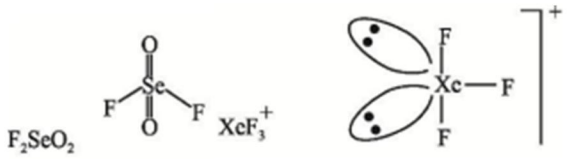
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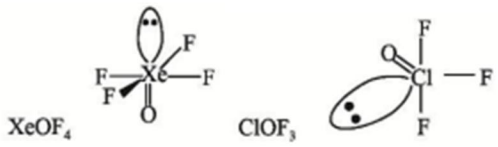
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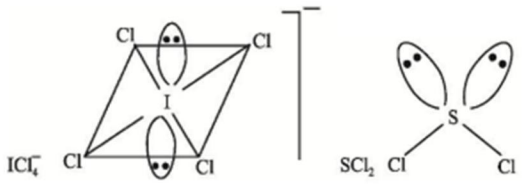
Species	Number of π^* electrons
O_2	2
O_2^-	3
O_2^{2-}	4
Total	= 9 electrons

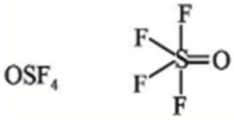












48. $E_{cell} = E_R - E_L$

$$0.516 = 0.28 - \frac{0.059}{2}\log\left[H^+\right]^2$$

$$= 0.28 + 0.059 \text{ pH}$$

$$\text{pH} = \frac{0.236}{0.059} = 4$$

49. Phosphorous acid (H_3PO_3) Dibasic $\therefore$ x = 2

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Tetrathionic acid ($H_2S_4O_6$) Dibasic $\therefore$ y = 2

Pyrophosphoric acid ($H_4P_2O_7$) tetrabasic $\therefore$ z=4

50. No of amino acids=4 and No of peptide bonds=3

MATHEMATICS

51.

$$I=\int e^x\left(\frac{1-x^{20}}{\left(1-x^{10}\right)\sqrt{1-x^{20}}}+\frac{10x^9}{\left(1-x^{10}\right)\sqrt{1-x^{20}}}\right)dx$$

$$=\int e^x\left(\sqrt{\frac{1+x^{10}}{1-x^{10}}}+\frac{10x^9}{\left(1-x^{10}\right)\sqrt{1-x^{20}}}\right)dx$$

$$=\int e^x\left(f(x)+f'(x)\right)dx=e^x f(x)+c=e^x\sqrt{\frac{1+x^{10}}{1-x^{10}}}+c$$

52.

$$\lim_{x\to\infty} n^{n^{-n}}\left((n+1)\left(n+\frac{1}{3}\right)\left(n+\frac{1}{3^2}\right)....\left(n+\frac{1}{3^{n-1}}\right)\right)^n$$

$$=\lim_{n\to\infty}\left(\frac{n+1}{n}\right)^n.\left(\frac{n+\frac{1}{3}}{n}\right)^n.....\left(\frac{n+\frac{1}{3^{n-1}}}{n}\right)^n$$

$$=\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n\left(1+\frac{1}{3n}\right)^{\frac{3n}{3}}.....\left(1+\frac{1}{3^{n-1}.n}\right)^{\frac{3^{n-1}.n}{3^{n-1}}}$$

$$=e.e^{\frac{1}{3}}.e^{\frac{1}{9}}.....e^{\frac{1}{3^{n-1}}}=e^{1+\frac{1}{3}+\frac{1}{3^2}.....\infty}$$

$$=e^{\frac{1}{1-1/3}}=e^{3/2}\quad;\qquad k=1.50$$

53. $\sin^2\beta=\cos\alpha.\sin\alpha$ and $\cos\alpha<\sin\beta<\sin\alpha$

$$\Delta=4\cot^2\beta-4$$

$$=4\left(\frac{\cos^2\beta-\sin^2\beta}{\sin^2\beta}\right)=4\left(\frac{1-2\sin^2\beta}{\sin^2\beta}\right)$$

$$=4\left(\frac{1-2\sin\alpha.\cos\alpha}{\sin^2\beta}\right)=4\left(\frac{\sin\alpha-\cos\alpha}{\sin\beta}\right)^2>0$$

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54.

$$\Delta = abc\begin{vmatrix} \frac{-bc}{a} & b+a & c+b \\ a+c & \frac{-ac}{b} & c+a \\ a+b & a+b & \frac{-ab}{c} \end{vmatrix} \quad ;$$

$$\Delta = \frac{abc}{abc}\begin{vmatrix} -bc & ab+ac & ac+ab \\ ab+bc & -ac & bc+ab \\ ac+bc & ac+bc & -ab \end{vmatrix}$$

$$\Delta = \begin{vmatrix} -bc & ab+ac & ac+ab \\ ab+bc & -ac & ab+bc \\ ac+bc & ac+bc & -ab \end{vmatrix} \quad ;$$

$$\Delta = \begin{vmatrix} -bc-ab-ac & 0 & ac+ab \\ ab+bc+ac & -ab-bc-ac & ab+bc \\ 0 & ab+bc+ac & -ab \end{vmatrix} \begin{matrix} C_1 \to C_1 - C_2 \\ C_2 \to C_2 - C_3 \end{matrix}$$

$$= (ab+bc+ac)^2 \begin{vmatrix} -1 & 0 & ac+ab \\ 1 & -1 & ab+bc \\ 0 & 1 & -ab \end{vmatrix} R_1 \to R_1 + R_2 + R_3$$

$$\Delta = (ab+bc+ac)^3 = 27$$

$$ab+bc+ac = 3 \; ; \quad a^2+c^2+b^2 = 2$$

$$a+b+c = \frac{-q}{p} \; ; \; (a+b+c)^2 = \frac{q^2}{p^2} \; ;$$

$$2+2\times 3 = \frac{q^2}{p^2} \Rightarrow \frac{q^2}{p^2} = 8 \; ;$$

$$\frac{q}{p} = \pm\sqrt{2}$$

But a + b + c > 0

So, $\frac{q}{q} < 0$

$$\frac{p}{q} = \frac{-1}{2\sqrt{2}} ; \; \frac{\sqrt{2}p}{q} + 1 = \frac{1}{2}$$

55. $P(G) \to$ Probability *that he guess the choice* $= 1/4$

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$P(C) \rightarrow$ Probability that he copies the choice = 1/8

$P(K) \rightarrow$ Probability that he knows the choice

$AC \rightarrow$ Event that answer is correct

$$P(G)=\frac{1}{4}, P(C)=\frac{1}{8}, P(K)=1-\frac{1}{4}-\frac{1}{8}=1-\frac{3}{8}=\frac{5}{8}$$

$$P\left(\frac{AC}{G}\right)=\frac{1}{5}, P\left(\frac{AC}{C}\right)=\frac{1}{10}, P\left(\frac{AC}{K}\right)=1$$

$$P\left(\frac{K}{AC}\right)=\frac{P(K \cap AC)}{P(AC)}$$

$$=\frac{\frac{5}{8}\times 1}{\frac{5}{8}\times 1+\frac{1}{4}\times\frac{1}{5}+\frac{1}{8}\times\frac{1}{10}}$$

$$=\frac{\frac{5}{8}}{\frac{5}{8}+\frac{1}{20}+\frac{1}{80}} \quad =\frac{\frac{5}{8}}{\frac{50+4+1}{80}}=\frac{50}{55}=\frac{10}{11}$$

56.

$$[\sin x+\cos x]=\left[\sqrt{2}\sin\left(x+\frac{\pi}{4}\right)\right]=\begin{cases}1 & 0\le x\le\frac{\pi}{2}\\ 0 & \frac{\pi}{2}<x\le\frac{3\pi}{4}\\ -1 & \frac{3\pi}{4}<x\le\pi\\ -2 & \pi<x\le\frac{3\pi}{2}\\ -1 & \frac{3\pi}{2}<x\le\frac{7\pi}{4}\\ 0 & \frac{7\pi}{4}<x\le 2\pi\end{cases} \text{ k}$$

$$=20\times\left[\frac{\pi}{2}-\pi+\frac{3\pi}{4}-2\times\frac{3\pi}{2}+2\pi-\frac{7\pi}{4}+\frac{3\pi}{2}\right]=20\times-\pi=-20\pi$$

57. Le the equation of sides AB, BC, CD and DA of parallelogram ABCD are respectively

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$y=\frac{3}{4}x+\frac{1}{4}$ ...(1); $y=\frac{3}{4}x+\frac{3}{4}$ ...(2)

$y=\frac{4}{3}x-\frac{1}{3}$ ...(3); $y=\frac{4}{3}x-\frac{2}{3}$ ...(4)

Here $m=\frac{3}{4}, n=\frac{4}{3}, a=\frac{1}{4}, b=\frac{3}{4}$,

$c=-\frac{1}{3}, d=-\frac{2}{3}$

$\therefore$ Area of parallelogram ABCD

$$=\left|\frac{(a-b)(c-d)}{m-n}\right|=\left|\frac{\left(\frac{1}{4}-\frac{3}{4}\right)\left(-\frac{1}{3}+\frac{2}{3}\right)}{\frac{3}{4}-\frac{4}{3}}\right|$$

$$=\left|\frac{-\frac{1}{2}\times\frac{1}{3}}{-\frac{7}{12}}\right|=\frac{2}{7}$$

58.

$$f'(x)=\begin{vmatrix}1 & 0 & 0\\ ab & x+b^2 & bc\\ ac & bc & x+c^2\end{vmatrix}+\begin{vmatrix}x+a^2 & ab & ac\\ 0 & 1 & 0\\ ac & bc & x+c^2\end{vmatrix}+\begin{vmatrix}x+a^2 & ab & ac\\ ab & x+b^2 & bc\\ 0 & 0 & 1\end{vmatrix}$$

$$=(x+b^2)(x+c^2)-b^2c^2+(x+a^2)(x+c^2)-a^2c^2+(x+a^2)(x+b^2)-a^2b^2$$

$$=3x^2+2x(a^2+b^2+c^2)<0$$

$f(x)$ is decreasing in

$$x\in\left(\frac{-2}{3}(a^2+b^2+c^2),0\right)$$

59.

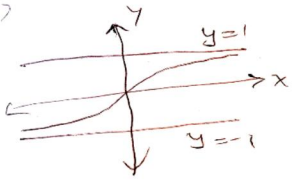
$$f(x)=\begin{cases}\frac{x}{1-x} & x<0\\ 0 & x=0\\ \frac{x}{1+x} & x>0\end{cases}$$

$$y=\frac{x}{1-x},\ x<0$$

$$x=\frac{y}{1+y}<0\Rightarrow y\in(-1,0)$$

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Also $y=\frac{x}{1+x},\ x>0 \Rightarrow x=\frac{y}{1-y}>0$

$y\in(0,1)$

So, range = (-1,1)

60.

$$\int\frac{dx}{x}=\int\frac{ydy}{1+y^2}$$

$$\ell nx=\frac{1}{2}\ell n\left(1+y^2\right)+c$$

Passes through (1, 0) so c = 0

$$x^2=1+y^2$$

$$x^2-y^2=1\rightarrow foci\rightarrow\left(\pm\sqrt{2},0\right)$$

Tangent at $\left(\sqrt{2},1\right)$

$$\sqrt{2}x-y=1$$

Equation of circle touching conic at $\left(\sqrt{2},1\right)$

$$\left(x-\sqrt{2}\right)^2+(y-1)^2+\lambda\left(\sqrt{2}x-y-1\right)=0$$

Passes through $\left(\sqrt{2},0\right)$ so

$$1+\lambda=0 \quad \Rightarrow \quad \lambda=-1$$

Equation of circle

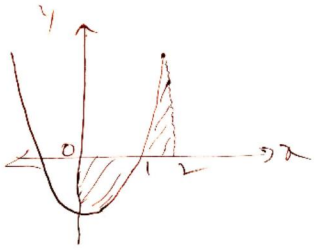
$$\left(x-\sqrt{2}\right)^2+(y-1)^2-\sqrt{2}x+y+1=0$$

$$x^2+y^2-3\sqrt{2}x-y+4=0$$

61. $A=\left|\int_0^1\left(x^2-1\right)dx\right|+\int_1^2\left(x^2-1\right)dx=2$

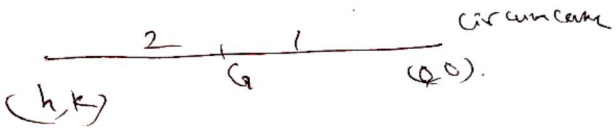
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62. $SS^1 = 2ae \quad sp + s^1 p = 2a$

63. Let $C(\cos\theta, \sin\theta) : H(h,k)$ is the orthocenter of the ΔABC



64.

Mean = $np = 3$

S.D. = $\sqrt{npq} = \frac{3}{2}$

$q = \frac{npq}{np} = \frac{9}{4\times 3} = \frac{3}{4}$

$p = 1 - \frac{3}{4} = \frac{1}{4}$

65. $T_2 = {}^n C_1 \left(a^{\frac{1}{13}}\right)^{n-1} \left(a\sqrt{a}\right) = 14a^{5/2}$

n = 14

66.

$a = A.(R)^{p-1} \rightarrow \log a = \log A + (p-1)\log R$

$b = A.(R)^{q-1} \rightarrow \log b = \log A + (q-1)\log R$

$c = A.(R)^{r-1} \rightarrow \log c = \log A + (r-1)\log R$

Angle between two vectors

$\vec{\alpha}.\vec{\beta} = 3\sum(q-r)\log a$

$= 0$

So, angle between vectors = $\frac{\pi}{2}$

67.

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$$\left|\vec{a}+\vec{b}\right|=\left|\vec{a}\right|+\left|\vec{b}\right|\Rightarrow \vec{a}\,||\,\vec{b}$$

$\therefore\ \lambda=1$

Hence $f'(1)=\frac{1}{2}\sin(1)$

68. $f(x)=y=x+\sin x$

$$\frac{dy}{dx}=1+\cos x$$

$$g'(y)=\frac{dx}{dy}=\frac{1}{1+\cos x}$$

Where $y=\frac{\pi}{4}+\frac{1}{\sqrt{2}}=x+\sin x$

$$\Rightarrow \qquad x=\frac{\pi}{4}$$

$$\therefore \qquad g'\left(\frac{\pi}{4}+\frac{1}{\sqrt{2}}\right)=\frac{1}{1+\left(\frac{1}{\sqrt{2}}\right)}$$

$$=\frac{\sqrt{2}}{\sqrt{2}+1}=\sqrt{2}\left(\sqrt{2}-1\right)$$

$$=2-\sqrt{2}$$

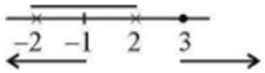
69.

$P\Rightarrow \quad x\in(-2,2)$

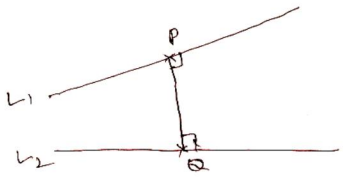
$Q\Rightarrow \quad x-1\geq 2$ or $x-1\leq -2$

$x\geq 3$ or $x\leq -1$

$P\cup Q=R-[2,3)$



70.



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$$L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$$

$$L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$$

Any point on

$L_1 : P(2r_1+1, 3r_1+2, 4r_1+3)$

$L_2 : P(3r_2+2, 4r_2+4, 5r_2+5)$

$\overline{PQ}$ perpendicular to the lines

$\therefore r_1 = \frac{1}{3} \qquad r_2 = -\frac{1}{6}$

$P\left(\frac{5}{3}, 3, \frac{13}{3}\right) \quad Q\left(\frac{3}{2}, \frac{10}{3}, \frac{25}{6}\right)$

71.

$$2^{2\log_5 4.\log_4 x + x\log_2 5} + 5^x + x^5 . x^{\log_5 4} + x^5 = 0$$

$$2^{2\log_5 x} \times 2^{x\log_2 5} + 5^x + x^5\left(4^{\log_5 x} + 1\right) = 0$$

$$5^x\left(2^{\log_5 x}\right)^2 + 5^x + x^5\left(\left(2^{\log_5 x}\right)^2 + 1\right) = 0$$

$$\left(5^x + x^5\right)\left(\underbrace{\left(2^{\log_5 x}\right)^2 + 1}_{\text{Always} > 0}\right) = 0$$

$5^x + x^5 = 0$ Only for $x < 0$

Only solution can be for $x < 0$

But for log to define $x > 0$

So, no solution exists

72.

$$z^2 - (3+i)z + (m+2i) = 0$$

Let one real root is k

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Then $k^2-(3+i)k+(m+2i)=0$

Real part = 0	imaginary part = 0
$k^2-3k+m=0$	$-k+2=0$
$m=2$	$k=2$

Let non real root is z_1

$2+z_1=3+i$

$z_1=3+i-2$

$z_1=1+i$

Product $=2\times1=2$

73.

$e^{x+y}=y^2$

$x+y=2\log y$

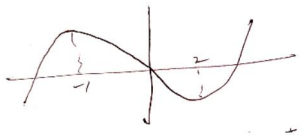
$1+\frac{dy}{dx}=\frac{2}{y}\frac{dy}{dx}$

$\frac{dy}{dx}=\frac{y}{2-y}$

$\frac{d^2y}{dx^2}=\frac{(2-y)\frac{dy}{dx}-y\times\frac{-dy}{dx}}{(2-y)^2}=\frac{2y}{(2-y)^3}$

$\left.\frac{d^2y}{dx^2}\right|_{x=-1}=2$

74. $f'(x)=0\Rightarrow x=2,-1$



$A=\int_{-1}^{0}\left(2x^3-3x^2-12x\right)dx+\int_{0}^{2}\left(12x+3x^2-2x^3\right)dx$

75.

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$$a = n!$$

$$b = \frac{(n-2)!}{(n-12)!}$$

$$c = (n-12)!$$

$$n(n-1) = 182$$

$$= 14 \times 13$$

$$n = 14$$

★★★★★ ALL THE BEST ★★★★★

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